## 1. Introduction

The sample switch is a critical component of the SAR ADC, responsible for accurately capturing the input signal and securing the sampled charge within the capacitor array. Each capacitor effectively interfaces with two switching elements: top-plate switches (the sample switch) and bottom-plate switches (MUX switches).

Because the sample switch is common to all top plates, it dominantly contributes to the total array's dynamic behavior compared to the MUX switches, which are distributed across multiple capacitor banks. Given that the positive and negative capacitor arrays are identical and feature two separate sample switches, this analysis focuses on sizing a single sample switch.

## 2. Design Considerations

Sizing the sample switch involves balancing several key performance metrics:

- **CDAC Settling Time:** Determines the maximum allowable on-resistance ($R_{on}$).
- **Switch Non-linearities:** The dependence of $R_{on}$ on the input voltage ($V_{in}$) and capacitor voltage ($V_{cap}$).
- **Holding Time & Charge Leakage:** Ensuring the captured charge does not degrade significantly during the conversion phase.
- **Charge Injection & Clock Feedthrough:** Minimizing errors introduced during the switch turn-off event.

## 3. Array Capacitance

Based on previous capacitor array sizing calculations, the unit capacitor ($C_u$) implementation was chosen to be 4.5 µm × 4.5 µm, yielding approximately 43.92 fF.

For an 8-bit architecture, the total capacitance per array ($C_{array}$) is:

$$C_{array} = 43.92 \text{ fF} \times 256 = 11.244 \text{ pF}$$

## 4. Settling Time and Maximum $R_{on}$ Criteria

To analyze the switch-on state, the input switch is modeled as a static resistor ($R_{on}$).

### 4.1 Worst-Case Voltage Step

Assume the maximum voltage step across the capacitor array is $V_{ref} = 0.9 \text{ V}$ (effectively $2V_{ref} = 1.8 \text{ V}$ differentially). While a full-scale step is highly unlikely during normal continuous tracking operation, designing for this worst-case scenario guarantees robust settling under all conditions. The voltage settling on the capacitor follows:

$$V_{cap}(t) = V_0 \left(1 - e^{-t/\tau}\right)$$

Where the time constant is $\tau = R_{on} \cdot C_{array}$.

### 4.2 Settling Time Target

We allocate 0.5 of a clock period for the capacitor settling time. For a relaxed 100 kHz clock, $T_{clk} = 10 \backslash \mathrm{mu\ s}$ (providing a tighter margin than the standard 96 kHz clock). The available settling time ($t_{settle}$) is **5 µs**.

### 4.3 Resolution Requirement

To ensure the sampled voltage represents the true input accurately, the settling error by the end of the sampling phase ($t_{settle}$) must be less than 0.5 LSB. The worst-case voltage error at time $t$ is:

$$V_{error}(t) = V_0 e^{-t/\tau}$$

We require $V_{error}(t_{settle}) < 0.5\ \mathrm{LSB}$. Since $1\ \mathrm{LSB} = \frac{V_{FS}}{2^N}$, $0.5\ \mathrm{LSB} = \frac{V_{FS}}{2^{N+1}}$. *(Note that $V_{FS}$ cancels out, making the required $R_{on}$ independent of the 0.9 V reference. $V_{FS}$= 2 $V_{ref}$ and $V_0 = V_{ref}$).*

$$\begin{aligned} V_0 e^{-t_{settle}/\tau} &< \frac{V_{FS}}{2^{N+1}} \\ e^{-t_{settle}/\tau} &< 2^{-(N)} \\ \frac{t_{settle}}{\tau} &> (N)\ln(2) \end{aligned}$$

*(Note: While $N\ln(2)$ is for differential case, rigorous 0.5 LSB settling for a worst-case full-scale step requires $(N+1)\ln(2)$ which can be seen in single ended cases where ( V_{FS} \) = ( V_{ref} \) . We will choose this criteria to favor accuracy.)*

$$R_{on} \leq \frac{t_{settle}}{C_{array} \cdot (N+1) \cdot \ln(2)}$$

### 4.4 $R_{on}$ Calculation

Substituting the physical design values ($t_{settle} = 5 \times 10^{-6}\ \mathrm{s}, N = 8, C_{array} = 11.244 \times 10^{-12}\ \mathrm{F}$):

$$R_{on} \leq \frac{5 \times 10^{-6}}{11.244 \times 10^{-12} \times (8+1) \times 0.693} \approx \mathbf{71.3\ k\Omega}$$

***Conclusion:*** *The upper bound for the sample switch on-resistance ($R_{on}$) is 71.3 kΩ. To account for PVT variations and switch non-linearities, the switch should be sized such that its peak resistance remains well below this threshold across all operating conditions. This is a very loose constraint for the $R_{on}$; practical simulations show that the actual $R_{on}$ is significantly lower across any process corner. (Note: Further analysis will be appended for charge injection, clock feedthrough, and leakage currents).*

## 5. Capacitor Array Leakage Current Analysis

### 5.1 Circuit Modeling

To analyze the leakage current, we model the capacitor array ($C_{array}$) with sampling switches and combined bottom-plate switches. For a single side of the differential array, the simplified model is:

- $V_{in}$ is connected through an "off" resistance ($r_{off}$) to the top plate of the capacitor.
- The capacitor $C_{array}$ is in series with another "off" resistance ($r_{off}$) connected to ground.
- $V_{cap}$ represents the voltage across the capacitor array.

### 5.2 Leakage Analysis Equations

For the leakage analysis, we use the fundamental charge equation:

$$Q_{cap} = C_{array} \cdot V_{cap}$$

For a small time interval ($\Delta t$), the change in charge relates to the change in voltage as follows:

$$\frac{\Delta Q_{cap}}{\Delta t} = C_{array} \frac{\Delta V_{cap}}{\Delta t}$$

The term $\frac{\Delta Q_{cap}}{\Delta t}$ represents the leakage current ($I_{leakage}$). Therefore:

$$I_{leakage} = C_{array} \frac{\Delta V_{cap}}{\Delta t}$$

### 5.3 Determining Worst-Case Parameters

To find the maximum allowable leakage current, we need the minimum allowable voltage drop ($\Delta V_{min}$) and the maximum hold time ($\Delta t_{max}$).

**Calculating $\Delta t_{max}$ (Hold Period):**

- **Design frequency (min):** 96 kHz
- **Clock Period ($t$):** $\frac{1}{96\ \text{kHz}} \approx 10.42\ \mu\text{s}$
- **Hold Periods:** The conversion takes multiple clock cycles (8 clocks for a single SAR conversion). To account for noise shaping and a safety margin, we assume the hold state lasts for 11 out of 12 clock cycles per conversion.
- **Calculation:** $11 \times 10.42\ \mu\text{s} = 114.62\ \mu\text{s}$
- **Result:** $\Delta t_{max} \approx \mathbf{115\ \mu s}$

### 5.4 Maximum Allowable Voltage Drop

The maximum allowable voltage drop per array is defined as half of the Least Significant Bit (LSB) voltage to maintain conversion accuracy:

$$\Delta V = \frac{V_{LSB}}{2} = \frac{2V_{ref}}{2^{N+1}} = \frac{0.9}{2^9}$$

*(Note: "0.9 / 2^9", is used for 8-bit resolution where the margin is applied at $N+1$ to account for errors introduced by both arrays.)*

$\Delta V_{min}$ **calculation:** $\frac{0.9}{512} \approx \mathbf{1.7578125\ mV}$

### 5.5 Maximum Leakage Current Calculation

Using the values derived above and a capacitor array value of $11.244\ \mathrm{pF}$:

$$I_{leakage_max} = 11.244\ \mathrm{pF} \times \frac{1.7578125\ \mathrm{mV}}{115\ \mu\mathrm{s}}$$

$$I_{leakage_max} = \mathbf{171.858\ pA}$$

**Design Limit:**
To ensure stability and accuracy, the allowable leakage must be less than the calculated limit:

$$I_{leakage_allowable} < I_{leakage_limit} = \mathbf{170\ pA}$$

### 5.6 Design Recommendations

By using this design limit, the following constraints are established for the system:

- Use capacitor arrays where $C_{array} > 11.244\ \mathrm{pF}$ (larger capacitance reduces the voltage droop for a fixed leakage).
- Maintain a clock frequency $f_{clk} > 96\ \mathrm{kHz}$ (higher frequencies shorten the hold time, reducing the impact of leakage).

## 6. Sample Switch FET Sizing Analysis

### 6.1 Design Strategy and Goals

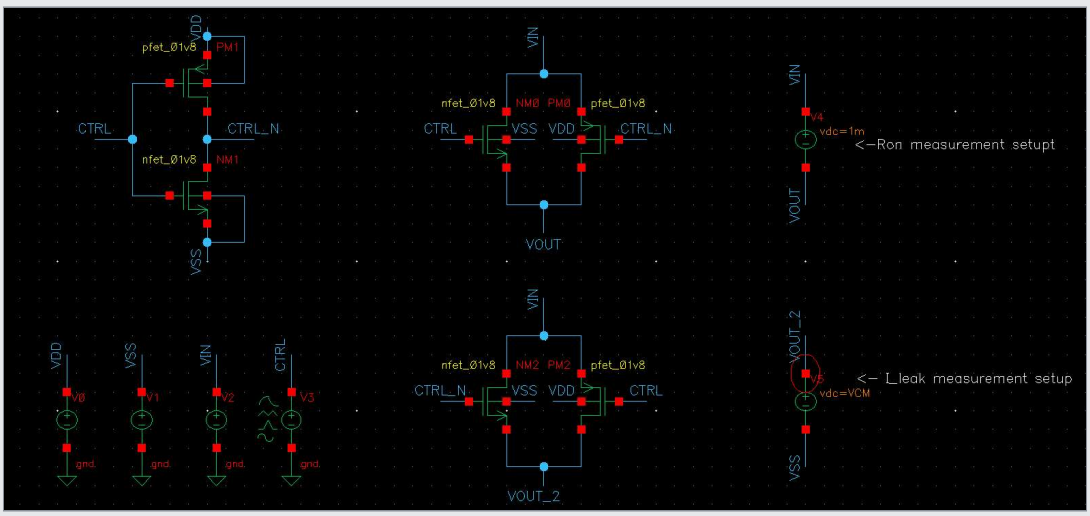
The sizing of the transmission gate (TG) necessitates a careful optimization between on-resistance ($R_{on}$) and leakage current ($I_{leak}$). To mitigate charge injection effects through first-order cancellation, the NMOS and PMOS devices are symmetrically sized ($W_n = W_p$ and $L_n = L_p$).

- **To minimize $R_{on}$:** Channel length ($L$) must be minimized while width ($W$) is maximized.
- **To minimize $I_{leak}$:** Channel length ($L$) must be increased while width ($W$) is minimized.

### 6.2 Simulation Methodology and Test Environment

Robustness is verified by simulating the design across worst-case Process, Voltage, and Temperature (PVT) corners:

- **Worst-Case $R_{on}$:** Evaluated at the SS (Slow-Slow) process corner at -25°C, where carrier mobility is lowest.
- **Worst-Case $I_{leak}$:** Evaluated at the FF (Fast-Fast) process corner at 125°C, where subthreshold and junction leakage currents peak.




***Description:*** *Cadence schematic illustrating the transmission gate setup with the capacitor array load and the DC/Parametric simulation sources.*

**Testbench Configuration:**

- $R_{on}$ **Characterization:** A 1 mV differential is forced across the TG input and output, and the resulting current is measured to calculate resistance.
- $I_{leak}$ **Characterization:** The TG output is biased at the common-mode voltage ($V_{CM}$), mirroring the SAR logic's tendency to drive the $C_{array}$ top-plate toward $V_{CM}$. The input voltage is swept to evaluate leakage.

**Simulation Strategy:**

Initial parametric sweeps of $W$ and $V_{in}$ across discrete lengths (150 nm to 220 nm) revealed edge-case vulnerabilities:

- $L = 150\ \mathrm{nm}$**:** Displayed excessive $I_{leak}$ at the upper limits of the input voltage range.
- $L = 220\ \mathrm{nm}$**:** Exhibited inadequate $R_{on}$ performance near the mid-rail input voltages.

Consequently, the methodology was refined: $V_{in}$ was fixed at the previously identified worst-case potentials (e.g., mid-rail for maximum $R_{on}$, and 0.05V / 1.75V for maximum $I_{leak}$, suppressing the absolute extremes to reflect realistic operational limits). $W$ and $L$ were then systematically swept to identify the optimal design boundary.

### 6.3 Parametric Sweep Results

The following tables summarize the design boundary conditions required to satisfy the established $R_{on}$ and $I_{leak}$ limits. Note that these values are approximations utilized to qualitatively guide the dimension selection process.

**Table 1:** $R_{on}$ **Boundary Constraints (SS Corner, -25°C)**
*Determines the maximum allowable $L$ for a given $W$ to satisfy the $R_{on}$ upper bound.*

| Width (W) | Length (L) Limit | Status |
|---|---|---|
| 3.0 µm | < 150 nm | ✗ (Fail) |
| 3.5 µm | < 154 nm | ✓ (Pass) |
| 4.0 µm | < 159 nm | ✓ (Pass) |
| 4.5 µm | < 164 nm | ✓ (Pass) |
| 5.0 µm | < 168 nm | ✓ (Pass) |
| 5.5 µm | < 176 nm | ✓ (Pass) |
| 6.0 µm | < 182 nm | ✓ (Pass) |
| 6.5 µm | < 190 nm | ✓ (Pass) |
| 7.0 µm | < 198 nm | ✓ (Pass) |
| 7.5 µm | < 205 nm | ✓ (Pass) |

**Table 2: $I_{leak}$ Boundary Constraints (FF Corner, 125°C)**
*Determines the minimum required $L$ for a given $W$ to strictly enforce the $I_{leak}$ upper bound.*

| Width (W) | Length (L) Limit | Status |
|---|---|---|
| 3.0 µm | > 224 nm | ✓ (Pass) |
| 3.5 µm | > 222 nm | ✓ (Pass) |
| 4.0 µm | > 221 nm | ✓ (Pass) |
| 4.5 µm | > 220 nm | ✓ (Pass) |
| 5.0 µm | > 220 nm | ✓ (Pass) |
| 5.5 µm | > 215.5 nm | ✓ (Pass) |
| 6.0 µm | > 211 nm | ✓ (Pass) |
| 6.5 µm | > 207.5 nm | ✓ (Pass) |
| 7.0 µm | > 204 nm | ✓ (Pass) |
| 7.5 µm | > 205 nm | ✓ (Pass) |

*Note: Extreme rail conditions ($V_{in} \approx 0\text{V}$ and $1.8\text{V}$) induce complex leakage mechanisms that fall outside the typical tracking envelope and were excluded from this specific boundary selection phase.*

### 6.4 Proposed Device Sizes

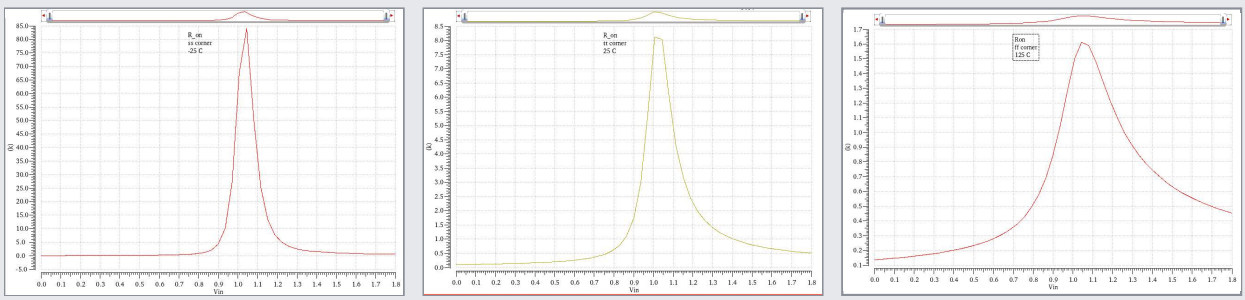
Evaluating the parametric sweep results to identify the minimum device footprint that simultaneously satisfies both the $R_{on}$ and $I_{leak}$ constraints across all worst-case corners yields the following optimal dimensions:

- $W_n = W_p = 7.5\ \mu\text{m}$
- $L_n = L_p = 205\ \text{nm}$

### 6.5 Final Simulation Verification

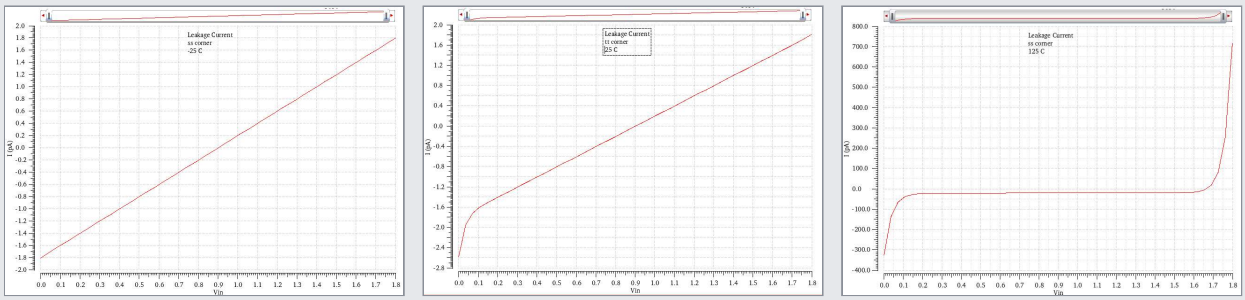
Deploying the proposed physical dimensions, the final verification simulations confirm the design viability:

- **On-Resistance ($R_{on}$):** Evaluated under the SS corner at -25°C, the constraint $R_{on} < 71.2\ \text{k}\Omega$ is strictly satisfied across the dominant input range, demonstrating significant headroom compared to the initial boundaries.



***Description:*** *Plots illustrating the $R_{on}$ curves across the 0V to 1.8V input span, capturing the peak resistance in the mid-rail region across different corners.*

- **Leakage Current ($I_{leak}$):** Evaluated under the FF corner at 125°C, the constraint $I_{leak} < 170\ \mathrm{pA}$ is strictly satisfied across the entire functional input range (excluding extreme rails).



***Description:*** *Plots illustrating leakage current versus input voltage, featuring a horizontal threshold marker at the 170 pA design limit across different corners.*

### 6.6 Design Rounding Considerations

If layout constraints necessitate rounding the channel length ($L$) to a standard grid, the dimensions can be adjusted to 200 nm or 210 nm:

- **Rounding to $L = 200\ \mathrm{nm}$:** Increases the leakage current ($I_{leak}$) significantly at the input voltage extremes.
- **Rounding to $L = 210\ \mathrm{nm}$:** Increases the peak on-resistance ($R_{on}$), particularly in the mid-rail region.

***Recommendation:*** *It is recommended to round up to $L = 210\ \mathrm{nm}$ while slightly increasing the device width (e.g., $W = 7.6\ \mu\mathrm{m}$). The reasoning is that the rate at which $R_{on}$ degrades is relatively lower compared to the exponential increase in $I_{leak}$ at shorter channel lengths. Furthermore, the established $R_{on}$ upper bound (71.3 kΩ) already contains significant redundancy, easily absorbing the minor resistance penalty introduced by this adjustment.*

## 7. Charge Injection and Switch Non-Linearities

### 7.1 Charge Injection Mitigation

To minimize the charge injection from the gate capacitance into the capacitor array ($C_{array}$) during the turn-off event, dummy switching devices are employed. Specifically, two dummy NMOS and two dummy PMOS devices are added alongside the active transmission gate devices. These dummy devices are sized with **half the width ($W/2$)** of the active devices and are driven by complementary clock signals to absorb the injected channel charge.

### 7.2 Switch Non-Linearities

To address switch non-linearities (such as the variation of $R_{on}$ with the input voltage), a bootstrapped switch architecture is often considered. However, in this design, a bootstrapped switch will **not** be utilized. The primary reasons are the inherent limitations of operating with low-voltage devices, alongside the significant circuit complexity and area overhead it would introduce. The symmetrically sized transmission gate, coupled with the conservatively established $R_{on}$ margins, provides sufficient linearity to satisfy the target 8-bit resolution.